

PPAR Audit

Explain why reported portfolio performance changed.

PPAR Audit answers a difficult operational question: "Why did my reported portfolio performance change?" It compares two portfolio-accounting snapshots, quantifies supported causes across holdings, transactions, prices, FX, and related data, flags suspicious source-data relationships, and produces reviewer-ready Excel and HTML reports. When the available evidence is insufficient, the unexplained difference stays visible for human review.

- Everything runs locally, so portfolio data stays inside your environment.
- The Python implementation supports automated batch runs and local customization.
- Standard output includes XLSX, HTML, CSV, JSON, and compact evidence bundles.

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What PPAR Audit Answers

PPAR Audit is built around one operational question:

Why did my reported performance change?

- **Performance Comparison:** identifies changed portfolio and security performance for each time period, quantitatively attributes defensible differences to supported source-data changes, and highlights anything that still needs human review.
- **Data Issues:** flags suspicious source-data relationships — including price ranges, dividend rates, accrued-interest rates, and missing dividends — that may indicate data-quality issues independently of the performance explanation.

Portfolio Audit Report

Executive Summary

PERFORMANCE DIFFERENCES

	Total Quantity	No Performance Differences	Fully Explained Differences	Partly Explained Differences	Unexplained Differences	Setup Incomplete
Portfolios	4	0	2	1	1	0
Portfolio Periods	30	4	24	1	1	0

DATA ISSUES

Issue Type	Quantity
holdings_price_range	20
large_price_variation	15
duplicate_transactions	4
missing_dividend	4
dividend_rate	3
holdings_accrued_rate	3
transactions_price_range	3
holdings_nonpositive_price	2
holdings_stale_price	2
pa_sa_rate	2
transaction_security_type_mismatch	2
transactions_nonpositive_price	2

Performance Differences

ROWS: 7

Portfolio	From Date	Thru Date	Performance Difference	Explained Difference	Unexplained Difference	Status	Comments
ALPHA	2026-01-31	2026-02-27	0.001368	0.001368		Fully Explained	
ALPHA	2026-05-01	2026-05-29	0.003571	0.003571		Fully Explained	
BALANCED	2026-04-01	2026-04-10	-0.002049	-0.002049		Fully Explained	
BALANCED	2026-05-09	2026-05-14	0.000149	-0.000051	0.000200	Partly Explained	Possible cause: csus.JPM transactions.amount increased by 200.00 on 2026-05-12. Add YAML configuration to count it as explained.
INCOME	2026-01-01	2026-01-30	-0.000060	-0.000060		Fully Explained	
INCOME	2026-02-28	2026-03-31	0.000002	0.000002		Fully Explained	
INCOME	2026-04-01	2026-04-30	0.000350	0.000000	0.000350	Unexplained	The Unexplained Difference may be due to missing source-data, source-file timing differences, or vendor methodology that does not match the YAML specifications.

Performance Difference Causes

ROWS: 29

Portfolio	From Date	Thru Date	As Of Date	Dataset.Field	Security	Snapshot A Value	Snapshot B Value	B - A Difference	Performance Difference Explained	Explanation
ALPHA	2026-01-31	2026-02-27	2026-01-30	holdings.accrued	fius91282Y5Y1	4.780000	4.830000	0.050000	0.000000	fius91282Y5Y1 beginning holdings.accrued increased by 0.05.
ALPHA	2026-01-31	2026-02-27	2026-01-30	holdings.market_value	causCASHUSD	11495.060000	9995.060000	-1500.000000	0.001421	causCASHUSD beginning holdings.market_value decreased by 1,500.00.
ALPHA	2026-01-31	2026-02-27	2026-02-10	transactions.amount	fius91282Y5Y1		-8.000000	-8.000000		pa: fius91282Y5Y1 transactions.amount decreased by 8.00. This affects the performance calculation through cash-balance ending holdings.market_value.
ALPHA	2026-01-31	2026-02-27	2026-02-17	transactions.amount	causCASHUSD		-2000.000000	-2000.000000	0.001965	lo: causCASHUSD transactions.amount decreased by 2,000.00. This affects the performance calculation through weighted external flow, which decreased by 785.71.
ALPHA	2026-01-31	2026-02-27	2026-02-27	holdings.market_value	causCASHUSD	7490.100000	5482.100000	-2008.000000	-0.002018	causCASHUSD ending holdings.market_value decreased by 2,008.00.
ALPHA	2026-01-31	2026-02-27	2026-02-27	holdings.quantity	causCASHUSD	7490.100000	5482.100000	-2008.000000		causCASHUSD ending holdings.quantity decreased by 2,008.00. This affects the performance calculation through holdings.market_value.
ALPHA	2026-05-01	2026-05-29	2026-04-30	holdings.market_value	causCASHUSD	-350.570000	-338.480000	12.090000	-0.000012	causCASHUSD beginning holdings.market_value increased by 12.09.
ALPHA	2026-05-01	2026-05-29	2026-05-29	holdings.market_value	causCASHUSD	-4355.530000	-2855.530000	1500.000000	0.001426	causCASHUSD ending holdings.market_value increased by 1,500.00.
ALPHA	2026-05-01	2026-05-29	2026-05-29	holdings.market_value	csusAAPL	226828.750000	229097.040000	2268.290000	0.002157	csusAAPL ending holdings.market_value increased by 2,268.29.
ALPHA	2026-05-01	2026-05-29	2026-05-29	holdings.price	csusAAPL	312.060000	315.180600	3.120600		csusAAPL ending holdings.price increased by 3.12. This affects the performance calculation through holdings.market_value.
ALPHA	2026-05-01	2026-05-29	2026-05-29	holdings.quantity	causCASHUSD	-4355.530000	-2855.530000	1500.000000		causCASHUSD ending holdings.quantity increased by 1,500.00. This affects the performance calculation through holdings.market_value.
BALANCED	2026-04-01	2026-04-10	2026-03-31	holdings.market_value	csusAAPL	156324.630000	157887.880000	1563.250000	-0.001732	csusAAPL beginning holdings.market_value increased by 1,563.25.
BALANCED	2026-04-01	2026-04-10	2026-03-31	holdings.market_value	csusJPM	31116.490000	31410.650000	294.160000	-0.000326	csusJPM beginning holdings.market_value increased by 294.16.
BALANCED	2026-04-01	2026-04-10	2026-04-06	transactions.amount	csusJPM	148.090000	158.670000	10.580000		dv: csusJPM transactions.amount increased by 10.58. This affects the performance calculation through cash-balance ending holdings.market_value.
BALANCED	2026-04-01	2026-04-10	2026-04-06	transactions.amount	csusJPM	-22.210000	-23.800000	-1.590000		dp: csusJPM transactions.amount decreased by 1.59. This affects the performance calculation through cash-balance ending holdings.market_value.
BALANCED	2026-04-01	2026-04-10	2026-04-10	holdings.market_value	causCASHUSD	54138.510000	54147.500000	8.990000	0.000009	causCASHUSD ending holdings.market_value increased by 8.99.
BALANCED	2026-04-01	2026-04-10	2026-04-10	holdings.quantity	causCASHUSD	54138.510000	54147.500000	8.990000		causCASHUSD ending holdings.quantity increased by 8.99. This affects the performance calculation through holdings.market_value.
BALANCED	2026-05-09	2026-05-14	2026-05-08	holdings.market_value	csusCVNA	293.810000	1469.050000	1175.240000	-0.001060	csusCVNA beginning holdings.market_value increased by 1,175.24.

BALANCED	2026-05-09	2026-05-14	2026-05-12	transactions.amount	csusJPM		200.000000	200.000000		rc: csusJPM transactions.amount increased by 200.00. This affects the performance calculation through cash-balance ending holdings.market_value. Add YAML configuration to count it as explained.
BALANCED	2026-05-09	2026-05-14	2026-05-14	holdings.market_value	csusMSFT	104657.590000	105797.530000	1139.940000	0.001009	csusMSFT ending holdings.market_value increased by 1,139.94.
BALANCED	2026-05-09	2026-05-14	2026-05-14	holdings.quantity	csusMSFT	255.617800	258.402000	2.784200		csusMSFT ending holdings.quantity increased by 2.7842. This affects the performance calculation through holdings.market_value.
INCOME	2026-01-01	2026-01-30	2026-01-20	transactions.amount	causCASHUSD	-50.000000	-100.000000	-50.000000		dp: causCASHUSD transactions.amount decreased by 50.00. This affects the performance calculation through cash-balance ending holdings.market_value.
INCOME	2026-01-01	2026-01-30	2026-01-22	transactions.amount	causMARGIN	-18.750000	-28.750000	-10.000000		ai: causMARGIN transactions.amount decreased by 10.00. This affects the performance calculation through cash-balance ending holdings.market_value.
INCOME	2026-01-01	2026-01-30	2026-01-30	holdings.market_value	causCASHUSD	123926.310000	123866.310000	-60.000000	-0.000060	causCASHUSD ending holdings.market_value decreased by 60.00.
INCOME	2026-01-01	2026-01-30	2026-01-30	holdings.quantity	causCASHUSD	123926.310000	123866.310000	-60.000000		causCASHUSD ending holdings.quantity decreased by 60.00. This affects the performance calculation through holdings.market_value.
INCOME	2026-02-28	2026-03-31	2026-02-27	holdings.accrued	fius91282Y5Y1	38.210000	38.400000	0.190000	0.000000	fius91282Y5Y1 beginning holdings.accrued increased by 0.19.
INCOME	2026-02-28	2026-03-31	2026-02-27	holdings.market_value	causCASHUSD	119921.350000	119677.960000	-243.390000	0.000247	causCASHUSD beginning holdings.market_value decreased by 243.39.
INCOME	2026-02-28	2026-03-31	2026-02-27	holdings.market_value	fius91282Y5Y1	48550.990000	48792.430000	241.440000	-0.000245	fius91282Y5Y1 beginning holdings.market_value increased by 241.44.
INCOME	2026-04-01	2026-04-30								No identifiable cause was found. Review `source_detail.csv`. The difference may be due to missing source-data, source-file timing differences, or vendor methodology that does not match the YAML specifications.

Data Issues

ROWS: 6

Snapshot	Portfolio	As Of Date	Dataset.Field	Security	Issue Type	Category	Reference Value	Observed Value	Difference	Tolerance	Explanation
Snapshot A	ALPHA	2026-05-23	transactions.amount	csusAAPL	missing_dividend	income				same-date dividend present	Missing a dividend for csusAAPL on 2026-05-23 that is in portfolio INCOME.
Snapshot A	INCOME	2026-01-30	holdings.price	csusGOOGL	holdings_price_range	price	313.000000	338.000000	25.000000	greater of 0.01 or 0.05%	Same-day same-security holdings.price differs across portfolios for csusGOOGL.
Snapshot B	ALPHA	2026-02-10	transactions.amount	fius91282Y5Y1	pa_sa_rate	accrued_interest	0.104377	0.159133	0.054757	greater of 0.000001 or 0.5%	pa: Same-day same-security rate differs across portfolios for fius91282Y5Y1.
Snapshot B	BALANCED	2026-04-06	transactions.amount	csusJPM	dividend_rate	income	1.399942	1.500012	0.100069	greater of 0.01 or 0.5%	dv: Same-day same-security rate differs across portfolios for csusJPM.
Snapshot B	INCOME	2026-01-30	holdings.accrued	fius91282Y5Y1	holdings_accrued_rate	accrued_interest	0.094983	0.096077	0.001094	greater of 0.00001 or 0.5%	Same-day same-security holdings.accrued per unit differs across portfolios for fius91282Y5Y1.
Snapshot	INCOME	2026-03-05	transactions.price	csusAAPL	transactions_price_range	price	260.290000	260.440000	0.150000	greater of	Same-day same-security

B

0.01 or 0.05% transactions.price differs across portfolios for csusAAPL.

Setup

PPAR requires Python 3.11.9 or newer.

Install the PPAR package:

```
pip install ppar
```

Create a local PPAR Audit workspace. The workspace includes PPAR-normalized demonstration data modeled on Axys/APX source and report data, so you can run the complete workflow before replacing the CSV files with reviewed exports from your own environment.

```
ppar setup ./my_ppar_audit
```

Run Audit:

```
ppar audit ./my_ppar_audit
```

Follow the `Customizing With Your Own Data` section in `./my_ppar_audit/README.md` when you are ready to customize the workspace with your own data.

```
my_ppar_audit/  
  README.md  
  ppar.yaml  
  run_audit.py  
  snapshot_a/  
    portperf.csv  
    holdings.csv
```

```
transactions.csv
secmast.csv
secperf.csv
splits.csv
snapshot_b/
  portperf.csv
  holdings.csv
transactions.csv
secmast.csv
secperf.csv
splits.csv
```

Inputs

The demonstration files are PPAR-normalized CSVs modeled on Axys/APX data. Production inputs may come from reviewed REP, IMEX, custom-report, or other controlled exports:

- portfolio performance;
- security performance;
- holdings;
- transactions;
- security master data;
- split factors.

Audit uses two source-data snapshots. Snapshot A is normally the older or original state and Snapshot B is normally the newer or restated state, but neither snapshot is presumed correct.

PPAR normalizes those files through a customizable `ppar.yaml` file, so each site can configure its local field names, transaction-code treatment, comparison tolerances, and report assumptions. The setup-created file documents these choices in place.

The configured file and accounting contracts fail closed when required source treatment is missing or ambiguous. Optional evidence does not silently expand the calculation or policy surface.

Outputs

PPAR Audit writes review packages:

```
output/  
  portfolio/  
    portfolio_audit.xlsx  
    portfolio_audit.html  
    source_detail.csv  
    audit_support.zip  
  security/  
    security_audit.xlsx  
    security_audit.html  
    source_detail.csv  
    audit_support.zip
```

To prevent unusably large artifacts, Audit stops with a nonzero exit code before writing a report when any primary review table would exceed 100,000 rows. The error identifies the oversized table and its largest contributors so the user can narrow the portfolio or date scope or correct upstream differences.

Current Validation Scope

PPAR Audit has substantial automated coverage, financial invariants, report reconciliation checks, output-integrity checks, deterministic demonstrations, and maintained scale gates.

It has not yet been validated against a real client's production-style Axys/APX exports and approved local accounting policy. The current program is seeking a small number of strong validation partners to test source authenticity, setup burden, financial interpretation, false positives, and reviewer usefulness.

PPAR Audit detects, compares, explains, and helps investigate supported portfolio-performance and source-data differences. It does not provide a financial-statement audit, GIPS verification, attestation, certification, or assurance opinion.

Additional Repository Capability

This repository also contains `ppar.analytics`, a maintained module for benchmark-relative performance attribution, contribution, and ex-post risk reporting. It is retained for future PPAR packaging but is not part of the current PPAR Audit validation program or default onboarding workflow.